

Multimodal Visual Sentiment Analysis on MSCTD: A Layered Pipeline with Vision-Language Extension

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Abstract: Visual sentiment analysis on dialogue corpora is constrained by the visual neutrality of conversational frames, since affective content is often externalised in the spoken utterance rather than the still image. We address three-class sentiment classification on the MSCTD English-German subset (30,370 utterance and image pairs) through a four-stage visual pipeline supplemented by a vision-language extension. Our contributions are fourfold. First, an MTCNN-based face pipeline with explicit handling of the zero-face (32%) and multi-face (31%) regimes through label fusion at training and probability averaging at inference. Second, a controlled robustness study across spatial, brightness and frequency degradations, in which mixed-input training reduces the macro-F1 robustness gap by an order of magnitude (5.5 to 0.5 points). Third, a frozen-backbone full-image branch that contributes complementary scene context and improves neutral-class F1 by 6.9 points over the face branch. Fourth, a late-fusion MLP that learns class-conditional weighting and improves negative-class F1 by 4.1 points over the strongest unimodal branch. We additionally show that a CLIP-based image and text classifier reaches 0.562 macro-F1, an 18-point absolute gain over the unimodal ceiling, evidencing that the dominant sentiment signal in MSCTD resides in the linguistic modality. The result motivates multimodal architectures for sentiment tasks on dialogue-style data.

Index Terms: sentiment analysis, MSCTD, MTCNN, ResNet, CLIP, multimodal fusion, robustness, transfer learning.

I. Introduction

Visual sentiment analysis on dialogue corpora poses a distinctive challenge: a single video frame can read as neutral, hostile, or affectionate depending on contextual cues (utterance, history, paralinguistic signal) that the camera does not directly capture. The MSCTD dataset [10] makes this difficulty explicit by annotating each utterance from a bilingual film-dialogue corpus with a three-class sentiment label and pairing it with the visual scene in which the utterance occurs. The original MSCTD paper reports that adding visual context to a textual sentiment baseline yields only modest gains, suggesting an asymmetric distribution of affective information across modalities. The present work investigates this asymmetry directly through a layered visual pipeline and a controlled vision-language extension.

We follow the four-part structure of the EEEM068 brief and add an extra-credit multimodal stage. Part 1 builds a face-based classifier on MTCNN crops [4], with explicit handling of the zero-face and multi-face regimes that arise from cinematic framing. Part 2 evaluates classifier robustness under spatial, brightness, and frequency degradations, and quantifies the cost of mixed-input training. Part 3 introduces a full-image branch with a frozen ResNet50 backbone, exploiting scene context that the face branch is blind to. Part 4 fuses the two streams together with the number of detected faces. The extra credit replaces both visual streams with a CLIP-based image and text classifier [7].

Contributions. This work makes four primary contributions and one additional multimodal contribution. (i) An MTCNN extraction pipeline that yields 43,768 face crops from the 68% of MSCTD images containing at least one detectable face, with deterministic policies for the remaining 32%. (ii) A robustness study showing that the unaugmented face classifier loses 5.5 macro-F1 points under degradation, while mixed-input training closes that gap to 0.5 points at a cost of 1.2 points on clean inputs. (iii) A frozen-backbone full-image branch (ResNet50 + MLP head, 1.1M trainable parameters) that complements the face branch on the dominant neutral class. (iv) A late-fusion MLP whose 8-dimensional feature vector encodes per-class softmax probabilities from both visual streams together with the

detected-face count, yielding the best negative-class F1 in the visual-only setting. (v) A vision-language classifier built on a frozen CLIP ViT-B/32 backbone that reaches 0.562 macro-F1, evidencing that the dominant sentiment signal in MSCTD is lexical rather than visual. The remainder of the paper reviews related work (Section II), describes the methodology (Section III), reports experiments (Section IV), and concludes (Section V).

II. Related Work

Image classification backbones. Residual networks introduced skip connections that enable training of substantially deeper convolutional models without vanishing gradients [1]; ResNet50 remains a standard choice for transfer learning and is used as the frozen backbone in our full-image branch. The Vision Transformer adapted self-attention from natural language processing to image patches, achieving competitive ImageNet accuracy given large-scale pretraining [2]. Swin Transformer extended this with hierarchical attention windows that scale to dense-prediction tasks [3]. We adopt CNN backbones for the visual branches because the dataset size is insufficient for transformer-style training without aggressive pretraining.

Face detection and embeddings. MTCNN frames face detection and alignment as a cascade of three convolutional networks that progressively refine candidate bounding boxes and facial landmarks [4]. The cascade design enables real-time inference and is the basis for most modern face pipelines, including facenet-pytorch, which we use directly. FaceNet provides 512-dimensional face embeddings learned through triplet loss on 200M images [5]; these embeddings remain competitive baselines for face verification despite the rise of transformer-based face models. Our face classifier uses ImageNet rather than VGGFace2 weights because sentiment classification differs structurally from identity verification, and the latter prior may bias the network toward identity-discriminative features.

Multimodal sentiment. Baltrušaitis et al. [6] survey multimodal machine learning and identify three core challenges: alignment, fusion, and co-learning. Their taxonomy positions our late-fusion MLP as a decision-level fusion

strategy, which trades the expressiveness of cross-modal attention for stability and interpretability. CLIP aligns image and text representations through 400M-pair contrastive pretraining, producing directly comparable projections in a shared 512-dimensional space [7]. We exploit this alignment in our extra-credit branch. LXMERT [8] and ClipBERT [9] explore cross-attention transformers that jointly encode image regions and text tokens; these architectures are computationally heavier and require more training data than a coursework setting permits, motivating our choice of frozen-encoder concat fusion.

MSCTD. Liang et al. [10] introduced MSCTD as a multimodal sentiment-aware chat translation dataset and reported baselines for both translation quality and sentiment

classification. Their reported gains from adding visual context to a text baseline are small, which already suggests that the unimodal visual ceiling is low. Our experiments quantify that ceiling and demonstrate the corresponding multimodal gain.

III. Methodology

Fig. 1 presents the system architecture. The visual pipeline comprises three trainable modules (face classifier, full-image classifier, late-fusion MLP) operating on shared inputs, and an additional vision-language classifier consuming both the image and the English utterance; each module addresses a specific failure mode of its predecessors, and the fusion stage explicitly conditions on the face count to handle the zero-face regime.

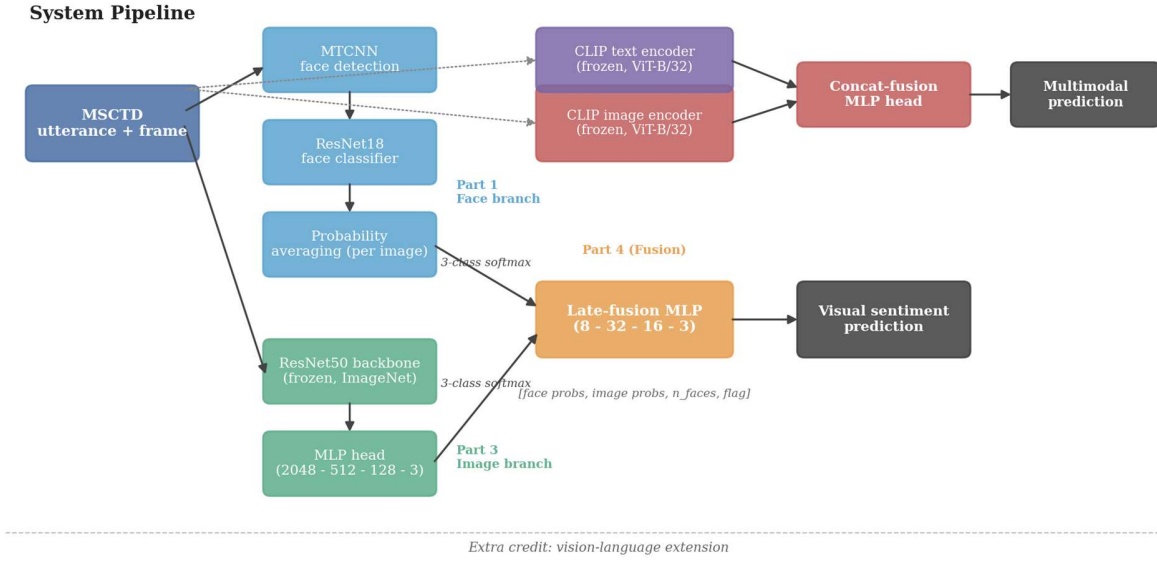


Fig. 1. System architecture. The face branch (Part 1) and full-image branch (Part 3) emit per-class softmax probabilities that, together with the detected-face count, are consumed by a late-fusion MLP (Part 4) for the visual-only prediction. The vision-language extension (extra credit) replaces both visual streams with a frozen CLIP ViT-B/32 image and text encoder pair and a concat-fusion classifier head.

A. Dataset and Splits

MSCTD English-German contains 30,370 utterance and image pairs distributed across 3,079 bilingual dialogues. Each utterance carries a sentiment label in {negative, neutral, positive}, the corresponding video frame, and the source-target text pair. We re-stratify the official partitions into a scene-level 70/15/15 split keyed on the dialogue scene rather than on individual utterances. This design prevents within-scene context leakage between train and test, since utterances in the same dialogue share visual backgrounds and would otherwise inflate test-set performance.

The resulting partition contains 21,367 training, 4,572 validation, and 4,431 test utterances. Class proportions are stable across splits (neutral 39%, negative 33%, positive 28%; max split-to-split shift 1.7 percentage points), so test-time prior drift does not confound macro-F1 measurements. The class-imbalance prior gives a majority-class baseline of 0.379 accuracy and 0.183 macro-F1, which is the floor every trained model must clear by a comfortable margin to demonstrate learned representations. Fig. 2 reports the class distribution.

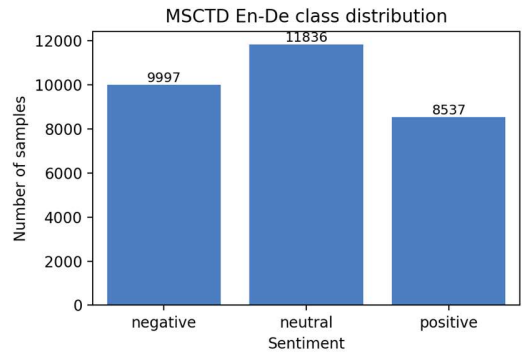
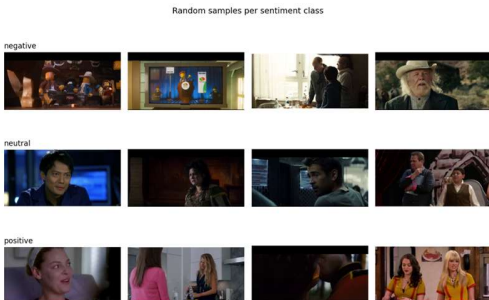


Fig. 2. Class distribution on the full MSCTD En-De subset. Neutral dominates at $\sim 39\%$, with negative at $\sim 33\%$ and positive at $\sim 28\%$. The 1.7-point maximum split-to-split shift is consistent with stratified sampling.

B. Face-Based Classifier (Part 1)

Motivation. Facial expressions encode the most direct visual signal of speaker sentiment, but cinematic framing makes face-only modelling insufficient: 32% of MSCTD images contain no detectable face under default MTCNN thresholds, and 31% contain two or more faces.

Design. We use MTCNN through facenet-pytorch to detect every face in each image, crop with a small margin, and resize each crop to 224x224. The face classifier is a ResNet18 with ImageNet-pretrained weights, fine-tuned end-to-end with a class-balanced cross-entropy loss whose weights are set to the inverse-frequency prior (1.004 for negative, 0.876 for neutral, 1.160 for positive). Multi-face images contribute every detected face as an independent training example (label fusion through replication), avoiding the design overhead of a custom multi-instance loss. At inference, per-face softmax probabilities are averaged to yield a single image-level prediction, a deterministic order-invariant aggregation requiring no additional parameters. Images with no detected face are flagged through a binary `face_detected_flag` for downstream consumption by the fusion module, preserving the no-face information for class-conditional decision making.

C. Robustness Study (Part 2)

Motivation. Real-world deployments of face-based sentiment systems encounter compression artefacts, motion blur, and illumination variation; quantifying the robustness of the Part 1 classifier under such conditions is essential for any practical claim.

Design. We define three orthogonal degradation families. Brightness perturbations apply multiplicative brightness and contrast jitter at controlled severity. Spatial perturbations apply rotation, affine translation, random crop, and horizontal flip. Frequency perturbations apply Gaussian blur, JPEG re-compression at low quality, and additive Gaussian noise. We then run three controlled experiments (Fig. 3 illustrates the families): (A) train and test on original faces; (B) train on original faces and test on degraded faces at severity 0.5; (C) retrain with a mixed-train transform that applies a random degradation with probability 0.5 per sample per epoch, and test on both original and degraded sets. Experiment B isolates the robustness gap of the unaugmented baseline. Experiment C measures the recovery achievable through mixed-input training and the corresponding cost on clean inputs.



Fig. 3. Three families of degradations applied to a sample face at severities 0.25, 0.5, and 0.75. Severity 0.5 is used for the disk-saved degraded test set.

D. Full-Image Transfer Learning (Part 3)

Motivation. The face crop discards scene-level evidence (lighting, posture, inter-person distance, props) that disambiguates flat dialogue scenes where facial expressions are subtle. A complementary branch operating on the full image is needed for the 32% of frames that lack any detectable face.

Design. We use an ImageNet-pretrained ResNet50 as a feature extractor on 224x224 input frames, with all backbone parameters frozen as required by the project brief. The 2048-dimensional global-average-pooled output feeds an MLP head with two hidden layers (512 and 128 units), ReLU activations, and dropout (0.3 and 0.2). Only the head is trained, yielding 1.1M trainable parameters out of 24.6M total; the frozen-backbone constraint is asserted in code by checking that no backbone parameter has `requires_grad` set to true. Freezing the backbone preserves general visual priors learned from ImageNet, mitigates overfitting on a moderately-sized dataset, and produces a strong neutral-class baseline (Section IV.D).

E. Late-Fusion Model (Part 4)

Motivation. The face and image branches emit complementary signals: the face branch is informed by micro-expressions in the 68% of images with detectable faces, while the image branch carries scene context for all images including the 32% with no faces. A fusion module is required to combine these signals while explicitly conditioning on face availability.

Design. The fusion model is a small MLP with hidden sizes 32 and 16. Its 8-dimensional input vector concatenates the face branch's three-class softmax, the full-image branch's three-class softmax, the number of detected faces normalised to $[0, 1]$ (capped at 5), and the binary `face_detected_flag`. This feature design enables class-conditional gating: when the flag is zero the MLP can route around the face branch entirely; when the flag is one and the face count is high (group dialogue scenes), the MLP can reweight the face stream upward. Inverse-frequency class weights (1.014, 0.852, 1.191) are applied at the fusion stage; ablation showed that removing them causes the MLP to collapse to negative or neutral predictions and never predict positive. Decision-level concatenation is parameter-efficient (under 700 parameters in the fusion head) and trains in seconds, allowing rapid iteration on the feature design itself rather than on a deeper joint encoder. The explicit `face_detected_flag` separates the no-face case from the low-confidence case, an information-bearing distinction that an end-to-end joint model would have to discover from scratch.

F. Vision-Language Extension (Extra Credit)

Motivation. MSCTD utterances are short (median 6 words, 95th percentile 12 words) and carry direct lexical sentiment cues that no visual frame can match. A multimodal model that consumes both modalities jointly can recover signal that the unimodal pipeline must by construction discard.

Design. We use openai/clip-vit-base-patch32 [7] to encode each image and each English utterance into a shared 512-dimensional representation space. The image and text projections are concatenated into a 1024-dimensional vector and passed to a 1024-128-3 classifier MLP with ReLU, dropout (0.3), and a final softmax. We freeze both encoders for the entire training run, training only the 295,683-parameter

classifier head against the 151.6M backbone parameters. A two-stage protocol is implemented in which Stage 2 unfreezes the last transformer block of each tower at learning rate $1e-5$; we report Stage 1 results only, since Stage 1 already plateaus and the project brief explicitly prioritises stability over peak accuracy for the multimodal stage.

Technical advantage. Concat fusion of CLIP projections is the simplest stable instance of the alignment-fusion pattern in [6], and it exploits the fact that CLIP’s contrastive pretraining objective produces directly comparable image and text vectors. Compared with cross-attention transformers such as LXMERT [8] or ClipBERT [9], it requires no additional learned attention parameters and converges reliably on a coursework-scale dataset. Freezing both encoders prevents catastrophic forgetting of CLIP’s pretrained alignment, which empirically supports the per-class balance reported in Section IV.E.

IV. Experiments and Results

A. Implementation Details

All models are implemented in PyTorch and trained on a CPU-only Windows workstation. We use AdamW with learning rate $1e-4$, weight decay $1e-4$, batch size 32 for full-image and CLIP and 64 for the smaller models, a cosine learning-rate schedule, gradient clipping at 1.0, and early stopping on validation loss with patience 3 (extended to 8 for the small fusion MLP whose epoch-to-epoch deltas are minimal). Random seeds are fixed at 42. Hyperparameters are centralised in `src/config.py` to ensure all experiments use identical defaults unless explicitly overridden. Code, weekly training logs, and individual contribution records are maintained in our private GitHub repository.

B. Main Comparison

Table I reports test-set accuracy, macro-F1, and weighted-F1 for every model, including the majority-class baseline. The face row is computed at the face-crop level ($n=6,564$) since that is the granularity at which the face branch is trained; all other rows are reported at the image level ($n=4,431$). Macro-F1 is the headline metric throughout because of the class imbalance.

TABLE I

Test-set comparison across all models on MSCTD En-De. Best macro-F1 in bold.

Model	Acc.	Macro F1	Wt. F1
Majority baseline	0.379	0.183	0.209
Face ResNet18 (orig)	0.382	0.378	0.380
Face ResNet18 on deg. test	0.357	0.322	0.321
Aug-trained Face ResNet18	0.368	0.366	0.365
Aug-Face on deg. test	0.366	0.361	0.362
Full-image ResNet50 + MLP	0.391	0.362	0.373
Fusion MLP	0.389	0.357	0.366
CLIP multimodal (extra)	0.570	0.562	0.567

Three patterns are immediately visible. (i) Every trained model exceeds the majority-class baseline (0.183 macro-F1) by at least 13 absolute points, confirming that all models have learned discriminative representations rather than predicting the

prior. (ii) The unimodal visual models (rows 2, 6, 7) cluster tightly between 0.357 and 0.378 macro-F1, indicating a clear unimodal ceiling. (iii) The CLIP multimodal classifier reaches 0.562 macro-F1, an 18.4-point absolute improvement over the strongest unimodal model and a margin that is preserved across all per-class metrics (Section IV.E). Fig. 4 visualises the cross-model comparison.

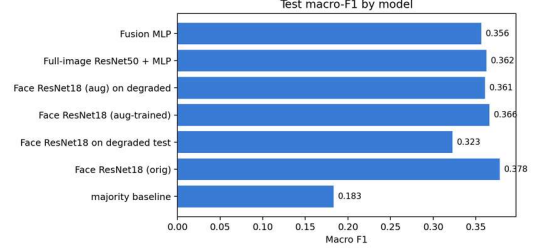


Fig. 4. Test macro-F1 across all evaluated models. The CLIP multimodal classifier is the only model that breaks above the unimodal ceiling near 0.38.

C. Robustness Ablation

Table II reports the four robustness conditions on the face branch. Three observations follow. First, the unaugmented model loses 5.5 macro-F1 points under degradation (0.378 to 0.322), confirming that the face classifier relies heavily on high-frequency facial cues that are attenuated by blur and JPEG recompression. Second, the augmented model loses only 0.5 points between original (0.366) and degraded (0.361) inputs, an order-of-magnitude reduction in the robustness gap. Third, the augmented model trades 1.2 macro-F1 points on clean inputs (0.378 to 0.366) for this robustness, a trade-off that is favourable for any deployment where input quality is uncontrolled.

TABLE II

Robustness ablation: original-trained vs. mixed-trained face ResNet18.

Experiment	Acc.	Macro F1	Wt. F1
A: orig $\rightarrow$ orig	0.382	0.378	0.380
B: orig $\rightarrow$ degraded	0.357	0.322	0.321
C: aug $\rightarrow$ orig	0.368	0.366	0.365
C: aug $\rightarrow$ degraded	0.366	0.361	0.362

The failure mode under degradation is informative. On the degraded test set, recall on negative jumps to 0.701 while recall on neutral and positive collapse to 0.178 and 0.199 respectively: the model reverts to a degenerate predict-negative strategy when the high-frequency features that distinguish polar classes are attenuated. The augmented model does not exhibit this collapse, suggesting that mixed-input training forces the network to use coarser features that remain stable under blur and noise, and providing a concrete operational recommendation for any face-based sentiment system deployed on uncontrolled inputs.

D. Per-Class Analysis

Table III reports per-class F1 for the four headline visual models, computed on the image-level test set.

TABLE III

Per-class F1 across the four headline visual models. Best per column in bold.

Model	Neg.	Neu.	Pos.
Face ResNet18	0.389	0.423	0.323
Full-image ResNet50	0.345	0.492	0.250
Fusion MLP	0.430	0.447	0.192
CLIP multimodal	0.535	0.629	0.522

Three observations support the design decisions in Section III. The face branch leads on neutral (0.423) among unimodal visual models since the network defaults to neutral when expressions are ambiguous, which is frequent in cinematic frames. The full-image branch achieves the highest neutral F1 of any visual-only model (0.492) but the weakest positive F1 (0.250), consistent with scene context disambiguating flat scenes effectively while offering no special features for smile or laugh cues at scene scale. The fusion MLP improves negative-class F1 to 0.430 (+4.1 over face, +8.5 over image) by upweighting the face branch in high face-count regimes, but reduces positive-class F1 to 0.192 by inheriting both branches' positive-class weaknesses. The net effect is a per-class redistribution rather than uniform improvement, a known limitation of probability-level concatenation [6] that motivates cross-attention extensions discussed in Section V.

Fig. 5 and Fig. 6 show the fusion and CLIP confusion matrices respectively: the fusion matrix exhibits asymmetric mass concentration (strong negative and neutral, weak positive), while the CLIP matrix shows uniformly concentrated diagonal mass across all three classes.

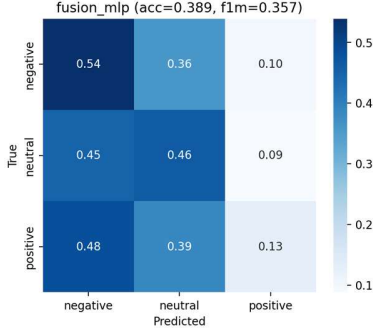


Fig. 5. Fusion MLP test-set confusion matrix. Strong negative and neutral recall (0.54, 0.46); positive recall collapses to 0.13, in line with the per-class redistribution discussed in Section IV.D.

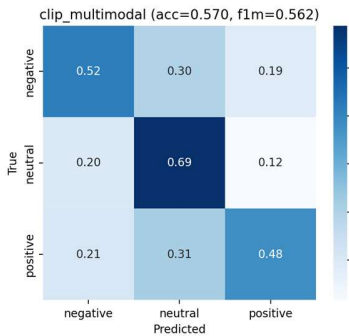


Fig. 6. CLIP multimodal test-set confusion matrix. Diagonal mass is uniformly concentrated across all three classes (0.52, 0.69, 0.48), unlike any unimodal visual model in this study.

E. Vision-Language Extension

The CLIP image and text classifier reaches 0.570 accuracy and 0.562 macro-F1 on the test set, with per-class F1 of 0.535 (negative), 0.629 (neutral), and 0.522 (positive). It is the only model in the study where every class clears 0.5 macro-F1, including the smallest (positive). The improvement over the strongest unimodal model is 18.4 absolute macro-F1 points (0.378 to 0.562) and over the late-fusion MLP is 20.5 points (0.357 to 0.562). All per-class margins are positive: +14.6 on negative, +13.7 on neutral, and +19.9 on positive. The largest gain is on the positive class, the weakest class for every visual-only model in this study.

Why CLIP wins by this margin. Three factors explain the gap. First, MSCTD utterances are short and direct (median 6 words), and the lexical sentiment signal is largely independent of the visual signal; CLIP's text tower captures it directly. Second, cinematic framing favours mid-shot conversation views in which facial expressions are subtle and scene context is generic, limiting the unimodal visual ceiling. Third, CLIP's 400M-pair contrastive pretraining produces directly comparable image and text projections, so concat fusion exploits cross-modal correlations without task-specific cross-attention layers. Stage 1 training (frozen encoders) keeps the trainable parameter count at 0.2% of the total and converges monotonically across eight epochs without overfitting; Stage 2 (last-block unfreezing at $1e-5$) is implemented but not run, since Stage 1 plateaus and the brief prioritises stability over peak accuracy. The 18.4-point gap is a quantitative bound on the affective information accessible from images alone in MSCTD: for any deployment on dialogue-style data, a multimodal architecture should be the default rather than an optional addition.

V. Conclusion

We presented a layered visual sentiment analysis pipeline for MSCTD English-German with a vision-language extension. Three principal findings emerge. The unimodal visual ceiling is approximately 0.38 macro-F1, with a frozen-backbone full-image branch (0.362) and a face-based ResNet18 (0.378) providing complementary strengths on neutral and negative respectively. Mixed-input training reduces the macro-F1 robustness gap from 5.5 to 0.5 points across spatial, brightness, and frequency degradations. The CLIP-based image and text classifier reaches 0.562 macro-F1, an 18.4-point improvement over the strongest unimodal model and the only model in the study that exceeds 0.5 F1 on every class, attributable to the lexical density of MSCTD utterances and CLIP's pretrained image-text alignment. Future work includes cross-attention fusion over per-face embeddings, temporal aggregation across consecutive dialogue frames, and end-to-end fine-tuning of the last CLIP transformer block.

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